

IT PROFESSIONAL PORTFOLIO

Cloud Engineering and Security Projects

PROJECT 25

Extend IAM Identity Center Administration with IAM Policy Management Permissions

AWS IAM | IAM Identity Center | Customer-Managed Policies | Permission Sets

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ENVIRONMENT	Personal AWS Lab
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STATUS	Complete

Executive Summary

Completed the IAM Identity Center skills challenge by creating a new customer-managed IAM policy named **ManagePoliciesForIdentityCenter** to allow trusted IAM administrators to view, create, update, and delete IAM policies. The initial policy included CreatePolicy, DeletePolicy, GetPolicy, and ListPolicies. After comparing against the solution, the missing policy-version and console dependency permissions were added: CreatePolicyVersion, DeletePolicyVersion, GetPolicyVersion, ListPolicyVersions, SetDefaultPolicyVersion, and ListEntitiesForPolicy. The completed policy was then attached to the IAM Administrators Identity Center permission set.

Business Problem

The IAM Administrators Identity Center persona could administer IAM Identity Center, but did not have enough IAM permissions to view, create, update, or delete customer-managed IAM policies. This creates an operational gap because Identity Center administrators must often create or attach policies to permission sets. The lab intentionally exposes that gap so the administrator can build a targeted policy rather than defaulting to broad AdministratorAccess.

Objectives

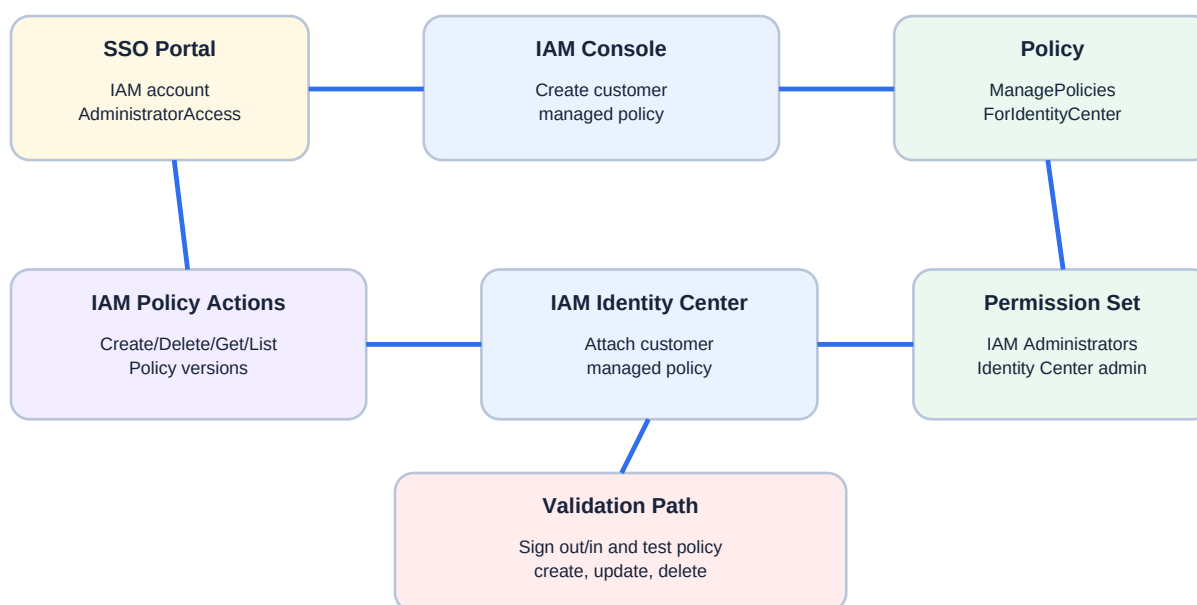
- Identify the missing IAM policy management capability in the Identity Center administration persona.
- Create a customer-managed IAM policy for IAM policy lifecycle operations.
- Start from a minimal self-directed action list, then compare against the solution.
- Add the missing policy version and console dependency actions.
- Name the policy ManagePoliciesForIdentityCenter.
- Attach the policy to the IAM Administrators Identity Center permission set.
- Document the difference between the initial attempt and the corrected final action list.

Environment

Primary Services	AWS IAM and AWS IAM Identity Center
Account Context	IAM account
Initial Access Path	IAM account with AdministratorAccess
Target Persona	IAM Administrators / Identity Center administration role
Policy Name	ManagePoliciesForIdentityCenter
Policy Resource Scope	*
Initial Actions	CreatePolicy, DeletePolicy, GetPolicy, ListPolicies

Corrected Actions Added	CreatePolicyVersion, DeletePolicyVersion, GetPolicyVersion, ListPolicyVersions, SetDefaultPolicyVersion, ListEntitiesForPolicy
Permission Set Updated	IAM Administrators Identity Center permission set
Purpose	Allow trusted Identity Center administrators to manage IAM policies needed by permission sets
Evidence Note	The portal screenshot shows both IdentityCenterAdmin and IdentityCenterAdministration roles available under the IAM account; the lab guide uses IdentityCenterAdministration naming.

Architecture



The customer-managed IAM policy fills the missing IAM policy management gap in the Identity Center administration persona.

Figure 1 - IAM policy management permission gap and corrected Identity Center administration path.

Implementation Summary

1. Opened the IAM account from the SSO access portal and used AdministratorAccess for the initial build.
2. Created a new IAM customer-managed policy without step-by-step direction as part of the skills challenge.
3. Initially allowed CreatePolicy, DeletePolicy, GetPolicy, and ListPolicies.
4. Compared the first attempt against the solution guide and identified missing API actions.
5. Added CreatePolicyVersion and DeletePolicyVersion to support modifying policies through policy versions.
6. Added GetPolicyVersion and ListPolicyVersions to support reading available policy versions.

7. Added SetDefaultPolicyVersion to allow the effective policy version to be changed.
8. Added ListEntitiesForPolicy so the console can check whether a policy is attached before deletion.
9. Set Resource to * because this policy is intended only for highly trusted IAM administrators.
10. Named the policy ManagePoliciesForIdentityCenter and added a descriptive purpose statement.
11. Attached the customer-managed policy to the IAM Administrators Identity Center permission set.

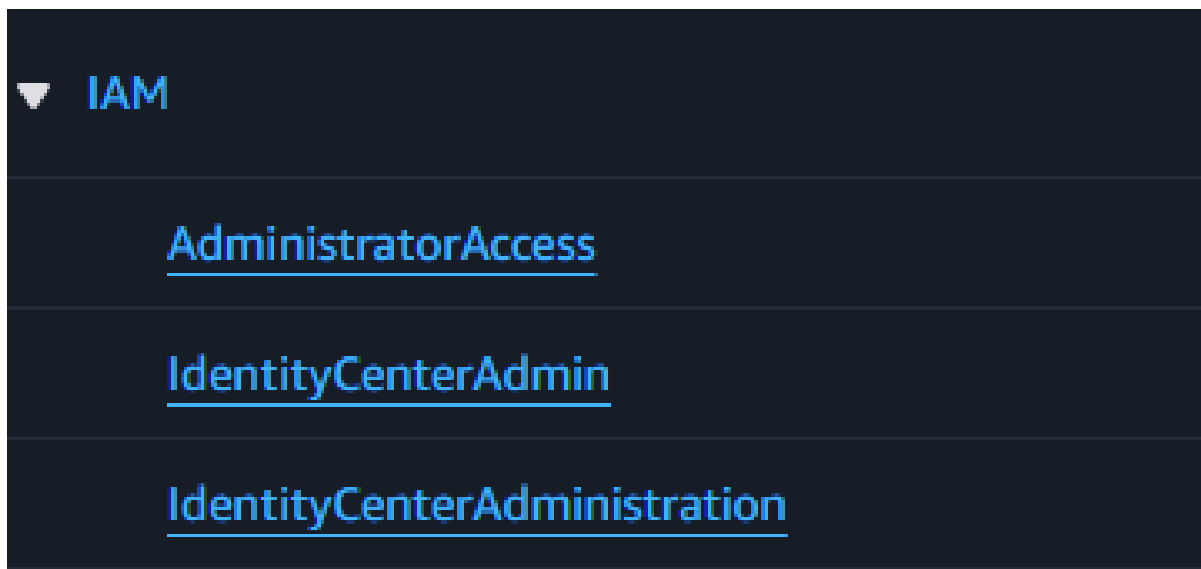
KEY LEARNING

The key learning point is that editing an IAM managed policy creates a new policy version. A user who can create a policy may still fail to update or delete policies unless the version-management and console-dependency actions are included.

Final Action List

CreatePolicy	Create policies	Allows new customer-managed policies to be created.
DeletePolicy	Delete policies	Allows cleanup of policies that are no longer needed.
GetPolicy / ListPolicies	Read policies	Allows the console and administrator to find and inspect policies.
CreatePolicyVersion	Update policies	IAM managed policy edits are implemented as new policy versions.
DeletePolicyVersion	Clean up versions	Allows removal of unneeded non-default versions.
GetPolicyVersion / ListPolicyVersions	Read versions	Allows version review before editing or setting defaults.
SetDefaultPolicyVersion	Choose active version	Controls which version is used when no version is specified.
ListEntitiesForPolicy	Console dependency	Lets the console check whether a policy is attached before delete.

Evidence

Figure 2 - IAM Account Shows Multiple Identity Center Administration Roles

The SSO access portal evidence shows the IAM account with AdministratorAccess, IdentityCenterAdmin, and IdentityCenterAdministration roles available. This supports the transition from full administrator access to a narrower Identity Center administration persona.

Figure 3 - Corrected JSON Policy Action List

The policy editor evidence shows the completed IAM policy action list including CreatePolicy, DeletePolicy, GetPolicy, GetPolicyVersion, CreatePolicyVersion, DeletePolicyVersion, ListPolicyVersions, ListPolicies, SetDefaultPolicyVersion, and ListEntitiesForPolicy with Resource set to *.

Figure 4 - Policy Details for ManagePoliciesForIdentityCenter

Review and create [Info](#)

Review the permissions, specify details, and tags.

Policy details

Policy name

Enter a meaningful name to identify this policy.

ManagePoliciesForIdentityCenter

Maximum 128 characters. Use alphanumeric and '+', '@', '-', '_' characters.

Description - optional

Add a short explanation for this policy.

Provides the Permissions to create, update, and delete policies in IAM Identity Center. Will work for any IAM

Maximum 1,000 characters. Use alphanumeric and '+', '@', '-', '_' characters.

Permissions defined in this policy [Info](#)

The policy review evidence shows the policy name ManagePoliciesForIdentityCenter and a description stating that it provides permissions to create, update, and delete policies for IAM Identity Center use.

Security Analysis

This policy is powerful because it allows policy lifecycle management. It should only be assigned to trusted IAM administrators and not to general security read-only users.

The policy fills a specific gap in the Identity Center administration persona instead of using AdministratorAccess for every Identity Center task.

Resource scope is wildcard because IAM policy management actions often require broad policy resources in the account. This makes group membership and permission set assignment the main risk boundary.

ListEntitiesForPolicy is a console-specific dependency that prevents delete workflows from failing unexpectedly when AWS checks whether a policy is attached to users, groups, or roles.

A production version should consider conditions, permission boundaries, change review, CloudTrail monitoring, and separation between policy authors and policy approvers.

Lessons Learned

IAM Identity Center permission sets can be extended with customer-managed policies, but those policies must exist in the target account with the exact expected policy name.

AWS console workflows may call extra APIs behind the scenes. Permission design should account for the entire workflow, not only the obvious final action.

IAM managed policy edits are version-based. Update permissions require CreatePolicyVersion, DeletePolicyVersion, GetPolicyVersion, ListPolicyVersions, and SetDefaultPolicyVersion.

Skills challenges are useful because they expose troubleshooting gaps that are easy to miss when following fully guided labs.

Production Improvements

- Restrict this policy to a dedicated IAM Administrators group with MFA and strong session controls.

- Monitor CreatePolicy, CreatePolicyVersion, SetDefaultPolicyVersion, DeletePolicyVersion, and DeletePolicy events through CloudTrail and Security Hub.
- Use change approval for policy changes in production.
- Use permission boundaries or guardrails to prevent identity administrators from creating privilege-escalation policies.
- Consider separate personas for policy authoring, policy approval, and break-glass administrator access.
- Test the Identity Center administration role after attaching the policy by creating, updating, and deleting a temporary low-risk policy.

Skills Demonstrated

IAM policies	IAM policy versions	IAM Identity Center
Customer-managed policies	Permission set extension	Console API dependencies
Least privilege tuning	Skills challenge validation	Role/persona testing

Resume Bullet

RESUME	Completed an AWS IAM Identity Center skills challenge by designing and correcting a customer-managed IAM policy for policy lifecycle operations, then attaching it to the IAM Administrators permission set to enable policy creation, updates, deletion, and console-based validation.
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STAR Interview Story

Situation: The IAM Identity Center administration persona could manage Identity Center but lacked IAM policy management permissions needed for real administration workflows.

Task: Create a targeted policy that allows IAM administrators to view, create, update, and delete IAM policies without relying on full AdministratorAccess for every session.

Action: I built an initial policy, compared it to the solution, added missing policy-version and console dependency actions, named it ManagePoliciesForIdentityCenter, and attached it to the IAM Administrators Identity Center permission set.

Result: The IAM administration persona now has the required IAM policy management capability while preserving a more intentional permission-set based access model.

Version History

v1.0.0	July 20, 2026	Initial documented implementation. Created ManagePoliciesForIdentityCenter, corrected missing IAM policy version and console dependency actions, attached the customer-managed policy to the IAM Administrators Identity Center permission set, and embedded provided evidence using portfolio documentation standard v1.0.1.

References

- Cloud Security Lab a Week (S.L.A.W.) - Skills Challenge: IAM Identity Center.
- Cloud Security Lab a Week (S.L.A.W.) - Skills Solution: IAM Identity Center.
- User-provided IAM SSO access portal role screenshot.
- User-provided IAM JSON policy editor screenshot.
- User-provided IAM policy review screenshot.
- Project 21 - Security Team Permission Sets in IAM Identity Center.